STAT 1210: Assignment 2

Part II: Probability and Distributions

Fall 2026 Semester

Instructions

- Your work should show all steps clearly for each question.
- Your solutions must be handwritten on lined or letter sized paper. Your name and student number must appear on the first page.
- Your name should also appear on every additional page and all pages should be stapled together.
- R is the preferred programming language, although equivalent Python code is also acceptable unless a question specifically requires R code.
- Programming resources, textbooks and software manuals may be consulted for coding tasks but all submitted work must be your own. Solutions generated by generative AI tools are not permitted.

Question 1

Let X be the number of seedlings, out of four, that show measurable root growth after a laboratory treatment. The proposed probability model is shown below.

x	0	1	2	3	4
$P(X = x)$	0.18	0.31	0.29	0.16	c

1. What value must c have for this to be a valid probability model and which probability rule determines that value?
2. What are $P(X \geq 2)$, $P(X < 3)$ and $P(X \neq 0)$?
3. What are the mean μ_X, variance σ_X^2, and standard deviation σ_X of X? Use the formulas below.

$$\mu_X = \sum_x xP(X = x)$$

$$\sigma_X^2 = \sum_x (x - \mu_X)^2 P(X = x)$$

$$\sigma_X = \sqrt{\sigma_X^2}$$

4. Verify your results using the following R code. Write the outputs of your code upto 6 decimal points.

```
> x <- 0:4
> p <- c(0.18, 0.31, 0.29, 0.16, 0.06)
> sum(p)
> mu <- sum(x * p)
> variance <- sum((x - mu)^2 * p)
> c(mean = mu, variance = variance, sd = sqrt(variance))
```

Question 2

The time T, in minutes, required for an automated instrument to complete a calibration is modelled as uniformly distributed between 2 and 8 minutes. The density is therefore constant over the interval $2 \leq T \leq 8$.

1. What height must the uniform density curve have so that its total area equals 1?
2. What are $P(3.5 \leq T \leq 6.2)$ and $P(T < 2.8 \text{ or } T > 7.4)$?
3. What is $P(T = 5)$ and why does this answer not mean that a calibration time of exactly 5 minutes is impossible in practice?
4. If the model is applied to 1200 independent calibrations, approximately how many calibration times would be expected to fall between 3.5 and 6.2 minutes? How does the Law of Large Numbers support this interpretation?

Question 3

A monitoring program classified 400 water samples by region and by whether the contaminant concentration exceeded a review threshold.

Region	Below threshold	Above threshold	Total
North	84	16	100
Central	126	24	150
South	90	60	150
Total	**300**	**100**	**400**

Let A denote "the sample is above the threshold" and S denote "the sample is from the South region."

1. What are the marginal probabilities $P(A)$ and $P(S)$? What is the joint probability $P(A \cap S)$?
2. What are $P(A \mid S)$ and $P(S \mid A)$? How do their denominators and interpretations differ?
3. What is the probability that a sample is either from the South region or above the threshold?
4. What are the conditional above-threshold rates within the North, Central and South regions? What evidence do these rates provide about an association between region and threshold status in this dataset?
5. Why would these observational data alone be insufficient to conclude that region causes the difference in threshold status?

Question 4

In a population of bacterial isolates, let A denote resistance to antibiotic A and B denote the presence of a particular plasmid marker. Suppose

$$P(A) = 0.32 \qquad P(B) = 0.25 \qquad P(A \cap B) = 0.12$$

1. Are A and B disjoint? What given information determines the answer?
2. What is $P(A \cup B)$?
3. What is the probability that an isolate has neither antibiotic A resistance nor the plasmid marker?
4. What are $P(A \mid B)$ and $P(B \mid A)$ and how should each probability be interpreted?
5. Are A and B independent?

Question 5

A laboratory obtains sterile containers from two suppliers. Supplier A provides 65% of the containers and has a 3% defect rate. Supplier B provides 35% of the containers and has an 8% defect rate. One container is selected at random from the combined inventory.

1. How should a complete probability tree for supplier and defect status be drawn and labelled? The outgoing branches from every node should sum to 1.
2. What are the four joint probabilities at the ends of the tree?
3. What is the overall probability that the selected container is defective?
4. Given that the selected container is defective, what is the probability that it came from Supplier B?
5. Given that the selected container is not defective, what is the probability that it came from Supplier A?

Question 6

A screening test is used to detect a plant pathogen. In the population being screened, 6% of plants carry the pathogen. The test has sensitivity 92% and specificity 95%. Assume these values apply to the population under consideration.

1. In a hypothetical group of 10000 plants, how many would be expected in each of the following categories: true positive, false negative, false positive and true negative?
2. What is the overall probability that a randomly selected plant receives a positive result?
3. Given a positive result, what is the probability that the plant carries the pathogen?
4. Given a negative result, what is the probability that the plant carries the pathogen?
5. Why is the sensitivity of 92% not the answer to Part 4? How does the 6% prevalence influence the interpretation of a positive result?

Question 7

Measurements from Laboratory A are approximately Normal with mean 120 units and standard deviation 15 units. Measurements from Laboratory B are approximately Normal with mean 80 units and standard deviation 5 units.

1. What are the z-scores of a value of 150 from Laboratory A and a value of 92 from Laboratory B?
2. Which measurement is farther above the centre of its own distribution? What feature of the z-scores supports the answer?
3. Under the Laboratory A model, what approximate proportion lies between 90 and 150 units according to the 68-95-99.7 rule?
4. Under the Laboratory A model, what are the exact probabilities $P(105 < X < 145)$ and $P(X > 150)$? Use the following R code to calculate these probabilities and write the output upto to 6 decimal places as your solution.

```
> pnorm(145, mean = 120, sd = 15) - pnorm(105, mean = 120, sd = 15)
> 1 - pnorm(150, mean = 120, sd = 15)
```

Question 8

The mass M of a packaged laboratory standard is modelled as approximately Normal with mean 250 g and standard deviation 3 g.

1. What two mass values enclose the central 90% of the model? The corresponding lower and upper cumulative probabilities should be identified before using `qnorm()`.
2. What mass marks the beginning of the heaviest 1% of packages?
3. In a production run of 2000 packages, approximately how many would be expected to exceed the cutoff from Part 2 if the model is accurate?
4. How should each cutoff be interpreted in the original units and context?

Question 9

For each situation below, which classification is most appropriate: **Binomial**, **Poisson** or **neither without additional assumptions**? For any situation classified as neither, what change or additional modelling condition might make a Binomial or Poisson approximation reasonable? Provide your reasons in 1-2 sentences.

1. A laboratory tests 20 independently selected samples. Each sample is classified as positive or negative, and the probability of a positive result is assumed to remain 0.08.
2. A researcher counts bacterial colonies appearing on equal-area plates when colonies occur independently at a stable average rate per plate.
3. Eight items are selected without replacement from a collection of 12 items containing five marked items, and the number of marked items selected is recorded.
4. A service records the number of requests received each hour but the average request rate changes substantially between daytime and nighttime.

Question 10

An optical sensor passes a calibration check with probability 0.85. Twelve sensors are checked under a model in which outcomes are independent and the pass probability is the same for each sensor. Let X be the number that pass.

1. Why does X satisfy the four Binomial conditions and how should its distribution be written?
2. What are $P(X = 10)$ and $P(X \geq 10)$? The first probability should be shown using the Binomial formula and both should be verified in R.
3. What are the mean and standard deviation of X?
4. How should the mean and standard deviation be interpreted across many repeated groups of 12 sensors?

Question 11

Small surface defects occur along a manufactured sheet at an average rate of 2.4 defects per 100 metres. Assume a stable rate and independent counts in non-overlapping lengths. Let Y be the number of defects in a randomly selected 250-metre length.

1. Why is a Poisson model appropriate and what value of λ applies to 250 metres?
2. What are $P(Y = 4)$, $P(Y \leq 2)$, and $P(Y \geq 1)$? (You may use R but show what commands you used)
3. What are the mean, variance and standard deviation of Y?
4. How would the value of λ change for a 50 metre length?

Bonus Question 1: Binomial Distribution

A quality study models the number X of components requiring adjustment among 80 independently produced components as Binomial. The adjustment probability p is constant, $p < 0.50$ and the variance of X is 12.8.

1. Using $\text{Var}(X) = np(1-p)$, what quadratic equation must p satisfy?
2. What are the two mathematical solutions for p, and which solution is consistent with the information $p < 0.50$?
3. What are the mean and standard deviation of X under the selected model?
4. What is $P(14 \leq X \leq 18)$? How can the inclusive interval be calculated using two cumulative Binomial probabilities?
5. What is $P(X \geq 25)$, and would a count of 25 or more be unusual under this model? The numerical result and the model assumptions should both be considered.

Bonus Question 2: Describing Probability

Choose one of the following options:

1. Describe two scientific probability related facts that you think I may not know. Explain each fact in at least 5 complete sentences.
2. In you own words, what does it mean to **count cards** at a casino? How would counting cards help with probabilities of winning a game of blackjack? Explain in 200-250 words.

You submission must be your own work. The answers will be assessed based on effort, originality and ability to explain probability in an understandable way. It will be not based on grammar ability.